College of Information Technology and Computing Department of Information Technology	**SYLLABUS** Course Title: IT Elective1 (Distributed Database Management Systems) Course Code: IT315 Credits: 3 units (2 hours Lecture, 3 hrs Laboratory)

USTP Vision

A nationally-recognized Science and Technology (S&T) university providing the vital link between education and the economy

USTP Mission

- Bring the world of work (industry) into the actual higher education and training of the students;
- Offer entrepreneurs of the opportunity to maximize their business potentials through a gamut of services from product conceptualization to commercialization;
- Contribute significantly to the national development goals of food security and energy sufficiency through technology solutions.

Semester/Year: 1st Semester / AY 2025-2026 Class Schedule: Th 1:00-4:00pm/W 3:00-5:00pm Th 1:00-4:00pm / T 10:00-12:00pm W 10:00-1:00pm / T 8:00-10:00am Th 7:00-10:00am / T 8:00-10am M 10:00-1:00pm / T 12:00-2:00pm Bldg./Rm. No. 09-304 / Dorm Classroom 1 09-305 / OC Dorm Classroom 1 / Dorm Classroom 2 09-305 / OC 09-302 / Dorm Classroom 2	Prerequisite(s): IT224, IT222, IT223 Co-requisite(s):
Instructor: PETAL MAY M. DAL Email: petalmay.dal@ustp.edu.ph Instructor: JHON HARVEY BABIA Email: jhonharvey.babia@ustp.edu.ph Instructor: GERALDINE BLANCO Email: geraldine.blanco@ustp.edu.ph	Consultation Schedule: W 1:00PM-3:00PM Bldg.Rm. No.: ICT Building (Building 9), 4th floor, IT Faculty Office Office Phone No./Local: 858-1738 local 1153 Consultation Schedule: F 1:00PM-5:00PM Bldg.Rm. No.: ICT Building (Building 9), 4th floor, IT Faculty Office Office Phone No./Local: 858-1738 local 1153 Consultation Schedule: M 1:00PM-5:00PM Bldg.Rm. No.: ICT Building (Building 9), 4th floor, IT Faculty Office Office Phone No./Local: 858-1738 local 1153

I. Course Description:

USTP Core Values:

A. **Unselfish Dedication** – Selfless commitment and complete fidelity towards a course of action or goal.
B. **Social Responsiveness** – Ethical/moral responsibility leading to corrective action on social issues and contributions for the betterment of the environment and the community's quality of life.
C. **Transformational Leadership** – Leading through inspiration and by example to foster positive change with the end goal of developing followers into leaders.
D. **Prudence** – Self-governance leading to circumspection and good judgment in the management of affairs and use of resources.

Program Educational Objectives:

PEO1: To develop the learner as a holistic person who is knowledgeable, productive and self-reliant member of the society with social, economic and environmental responsibility.

PEO2: To equip students with appropriate knowledge and skills, proper attitudes and values towards work.

PEO3:

The first is an in-depth study of the classical distributed database management issues such as distribution design, distributed query processing and optimization, and distributed transaction management. The second objective is to study more current distributed database management topics such as pervasive computing, Web data management, different distribution models (push versus pull), interoperability and componentization.

II. Course Outcome:

Course Outcomes (CO)	Program Outcome (PO)														
	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
CO1: Explain the architecture, advantages, limitations, and key differences between centralized and distributed database systems.	I	I	I												
CO2: Design and evaluate data distribution strategies (e.g., fragmentation and replication) for distributed databases based on system requirements.			E	E	E										
CO3: Implement and test distributed database solutions using NoSQL technologies, applying key concepts like consistency, availability, and partitioning.					D	D	D								
CO4: Apply distributed query optimization and transaction management techniques to ensure reliable and efficient data operations in a distributed environment.		D	D				D								

III. Course Outline:

To translate vocational interest and diversified occupational skills through the effective utilization of appropriate technology for people's economic sufficiency.

Program Outcomes:

a: Apply knowledge of computing, science, and mathematics in solving computing/IT-related problems through critical and creative thinking.

b: Use current best practices and standards in solving complex computing/IT-related problems and requirements;

c: Analyze complex computing/IT-related problems by applying analytical and quantitative reasoning; and define the computing requirements appropriate to its solution;

d: Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer based-systems;

e: Design creatively, implement and evaluate different computer-based systems, processes, components,

Allotted Time	Course Outcomes (CO)	Intended Learning Outcomes (ILO)	Topic/s	Suggested Readings	Teaching-Learning Activities	Assessment Tasks/Tools	Grading Criteria	Remarks
Week 1 (Aug. 7-8, 2025) 2 hrs			Course Orientation (Class Policies & requirements) Orientation on the USTeP portal Creation of online student account.	*Student Handbook *Course Syllabus	Discussion			
Week 1-2 (Aug. 7-14, 2025) 8 hrs	CO1	1. Define distributed database concepts. 2. Differentiate centralized vs distributed systems. 3. Explain CAP and PACELC theorems and their implications.	Module 1: Introduction to Distributed Management System and CAP & PACELC Theorems	IT315 DDBMS IM	Flipped Classroom	Quiz		

or programs to meet desired needs and requirements under various constraints; f: Integrate effectively the IT-based solutions into the user environment with appropriate consideration for public health and safety, cultural, societal, and environmental concerns; g: Select, adapt, and apply appropriate techniques, resources, skills and modern computing tools to complex computing activities, with an understanding of the limitation; h: Function effectively as individual, or work collaboratively and respectfully as a member or a leader in diverse development and/or multicultural settings; i: Assist in the creation of an effective IT project plan; j: Communicate effectively in both 0oral and in written form by being able to deliver and comprehend instructions clearly; and present persuasively to diverse audience the complex computing / IT-related ideas and perspectives; k: Assess local and global impact of computing information technology on individuals. Organizations, and society;	Week 2-4 (Aug. 11-29) 10 hrs	CO2	1. Identify fragmentation types: horizontal, vertical, hybrid. 2. Evaluate replication strategies for availability and fault tolerance. 3. Match distribution methods to system requirements.	Module 2: Data Distribution Alternatives	IT315 DDBMS IM *** Distributed Database Management System : A Practical Approach** Saeed K. Rahimi and Frank S. Haug *** Principle of Distributed Database System 3rd Edition** M.Tamer Ozsu, Patrick Valduriez https://www.tutorialspoint.com/distributed_dbms/distributed_dbms_databases.htm https://cassandra.apache.org/doc/latest/	Demonstration	Laboratory Activity Quiz			

l: Act in recognition of professional, ethical, legal, security and social responsibilities in the utilization of information technology; m: Recognize the need to engage in independent learning and be at peace with the latest developments in a specialized field in IT, with emphasis on Database Management and Information System; Network Design and Administration; and Computer Vision and Image processing for continual development as a computing professional; n: Participate in generation of new knowledge; or in research and development projects aligned to local and national development agenda or goals with the end view of contributing to the local and national economy; and o: Preserve and Promote “Filipino historical and cultural heritage.”	Week 4-6 (Aug. 25-Sept. 5, 2025) 10 hrs	CO3	1. Explain NoSQL data models (key-value, column-family, document, graph). 2. Contrast ACID and BASE properties. 3. Apply CAP theorem to NoSQL database design.	Module 3: NoSQL Models & ACID vs BASE	IT315 DDBMS IM *** Distributed Database Management System : A Practical Approach** Saeed K. Rahimi and Frank S. Haug *** Principle of Distributed Database System 3rd Edition** M.Tamer Ozsu, Patrick Valduriez	Flipped Classroom Demonstration	Laboratory Activity Hands-on quiz		
	Week 6-8 (Sept. 8-26, 2025) 10hrs	CO3	1. Define and design keyspaces, tables, and columns in NoSQL. 2. Create a basic NoSQL schema.	Module 4: Data Models	IT315 DDBMS IM *** Distributed Database Management System : A Practical Approach**		Laboratory Activity Hands-on quiz		

			3. Validate design based on access patterns.		Saeed K. Rahimi and Frank S. Haug * **Principle of Distributed Database System 3rd Edition** M.Tamer Ozsu, Patrick Valduriez				
	MIDTERM EXAMINATION								
	Week 10-11 (OCt. 9-17, 2025) 10 hrs	CO3	1. Use Cassandra Query Language (CQL) syntax. 2. Perform CRUD operations in a distributed NoSQL setup. 3. Test queries for efficiency and correctness.	Module 5: Query Language	IT315 DDBMS IM https://cassandra.apache.org/doc/latest/	Jigsaw Creative Presentation	Laboratory Activity Hands-on quiz		

	Week 12-13 (Oct. 20-31, 2025) 10 hrs	CO4	1. Implement indexing in NoSQL for performance. 2. Perform aggregation queries. 3. Optimize queries in a distributed environment.	Module 6: Indexing, Aggregation, and Query Optimization	IT315 DDBMS IM https://www.tutorialspoint.com/distributed_dbms/distributed_dbms_databases.htm https://cassandra.apache.org/doc/latest/	Demonstration Quiz Bee	Laboratory Activity Quiz			
	Week 14-15 (Nov. 10-21, 2025) 10 hrs	CO4	1. Explain distributed transaction management concepts. 2. Apply concurrency control methods. 3. Test consistency during concurrent transactions.	Module 7: Transaction and Concurrency Control	IT315 DDBMS IM https://www.tutorialspoint.com/distributed_dbms/distributed_dbms_databases.htm https://cassandra.apache.org/doc/latest/	Demonstration	Laboratory Activity Quiz			

Week 16-17 (Nov. 24-Dec. 1, 2025) 10 hrs	CO4	1. Perform backup operations in NoSQL. 2. Restore databases to a consistent state. 3. Verify data integrity after recovery.	Module 8: Backup and Restore	IT315 DDBMS IM https://www.tutorialspoint.com/distributed_dbms/distributed_dbms_databases.htm https://cassandra.apache.org/doc/latest/	Brainstorming	Laboratory Activity Quiz		
FINAL EXAMINATION								

IV. Course Requirements:

1. Class standing (attendance, participation, etc.) policy:

(a) Expected classroom behavior (may want to develop this with the students, e.g., What guidelines m are appropriate for behavior and participation in a large class

- Students must come to class on time.
- Strict observance of deadlines.
- Class participation is encouraged.
- Observe proper courtesy.

(b) Ground Rules for participation in discussions or activities.

- Only one student may talk at a time.
- Must follow instructions for every activity given.
- For group activity, each member must participate accordingly.

2. Course Readings/Materials:
   (a) Titles, authors, and editions of textbooks and other materials, required and recommended

   1. IT315 – Distributed Database Management System Instructional Material by: Petal May M. Dal

   2. Distributed Database Management System : A Practical Approach
   Saeed K. Rahimi and Frank S. Haug
   ISBN 978-0-470-40745-5

   3. Principle of Distributed Database System 3rd Edition
   M.Tamer Ozsu, Patrick Valduriez

   (b) Supplies needed (calculators, software, workbooks, disks, CDs, lab supplies, art supplies, etc.)
   - Apache Cassandra

   (c) URLs for online resources
   - https://cassandra.apache.org/doc/latest/
   - www.couchbase.com
   - www.techtarget.com

3. Assignments, Assessment, and Evaluation
   (a) Policy concerning homework (grading, posting, late policy, etc.)
   Students may share ideas as they work on their assignments but the submitted assignments must be their own work.
   (b) Policy concerning make-up exams
   No special examination is given unless a student has valid reasons stipulated in the Student Handbook Article 3: Excused Absences.
   (c) Policy concerning late assignments/requirements
   - Assignments: no assignment for a particular date, will have a grade of zero (0).
   - Projects: late submission of projects will have a corresponding consequence. There will be a deduction of points for every day that the project submission will be late.

   (d) Preliminary information on term papers or projects, with due dates
   - Projects for midterm and finals are given ahead of time along with its corresponding due dates,

rubrics, and other requirements for the completion of the projects.

- Rubrics for PIT

Member Names: ____________________
____
____
____

PIT Rubric

criteria	outstanding 10pts	very satisfactory 8pts	satisfactory 7pts	needs improvement 5pts	score
data model	well-defined data model	85% of data model can be understood	70% of data model can be understood	<70% of data model can be understood	
all keyspaces are defined and populated accordingly	all keyspaces are defined and populated	85% of keyspaces are defined and populated	70% of keyspaces are defined and populated	<70% of keyspaces are defined and populated	
distribution strategy	either simple or network			no distribution strategy	
implemented CQL queries	all queries are functioning	85% or queries are working properly	70% or queries are working properly	<70% or queries are working properly	
				total	

- Non-submission of projects does not mean you

(e) List of assignments that will impact the final grade and % weight given each

- Portfolio: grade will be part of the PIT.

(f) Description in detail of grading processes and criteria (how many quizzes, tests, papers; weighting of each; amount of homework, etc.) or the GRADING POLICY

Grading System

Lecture Grade (67%)	
Performance Item/Criteria	**%**
Class Performance Item	10%
Quizzes (All quizzes, prelim and pre-final exams)	40%
Major Exams (i.e, Midterm and Final Exams)	30%
Performance Innovative Task / Project	20%
TOTAL	**100%**
Laboratory Grade (33%)	
Performance Item/Criteria	**%**
Laboratory Exercises/Reports	30%
Laboratory Major Exam	40%
Hands on Exercises	30%
TOTAL	**100%**
Term/Periodic Grade = 67% Lecture Grade + 33% Laboratory Grade	

Options:
FINAL GRADE (FG) = 1/3 Midterm Grade (MTG)+ 2/3 Final Term Grade (FTG)
FINAL GRADE (FG) = 1/2 Midterm Grade (MTG)+ 1/2 Final Term Grade (FTG)

(Passing Percentage is 70%)
Ex. In a 10-item quiz, obtaining 7 points would be equivalent to a passing score.

Disclaimer:
Every attempt is made to provide a complete syllabus that provides an accurate overview of the subject. However, circumstances and events make it necessary for the instructor to modify the syllabus during the semester. This may depend, in part, on the progress, needs, and experiences of the student.

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